IEE1860 BioMEMS
Lab2: FEM I. - basics

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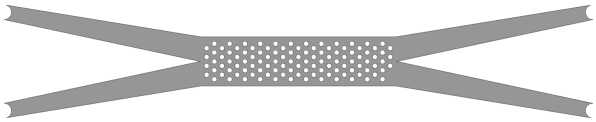
1. Objectives

The primary objective of this lab is to use FEM simulation of fluid flow and species transport in the microfluidic reactor we designed in the previous lab. The secondary objective is to familiarize myself with the simulation software Comsol.

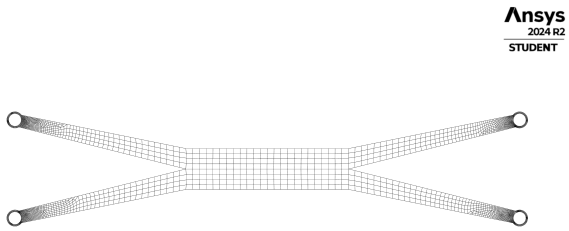
2. Methodology

Since my home institution is using Ansys for FEM analysis I decided to deviate from objectives and use this software. With this, I could use computers I have access to and didn't have to use VPN to TalTech lab. After this decision, I used Ansys SpaceClaim tool to convert the exported face from Autodesk Inventor to geometries and Ansys workbench meshing tool to generate meshes. The resulting geometries and meshes are shown in Figure 1.

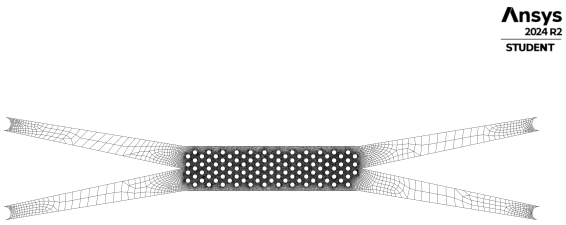
(a) Geometry without micropillars



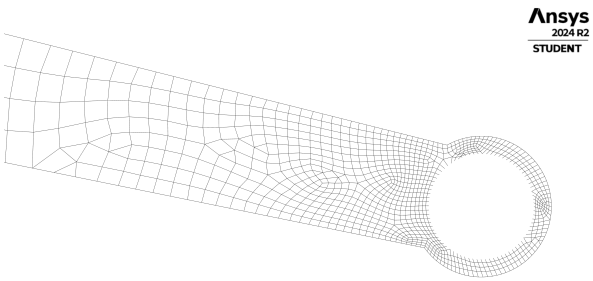
(b) Geometry with micropillars



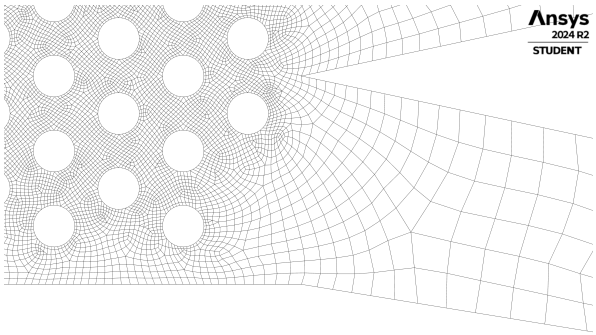
(c) Mesh without micropillars



(d) Mesh with micropillars



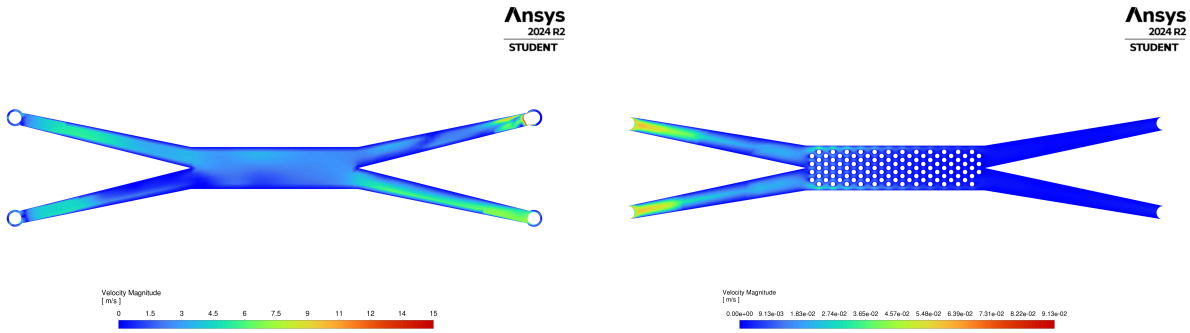
(e) Mesh around outlet



(f) Mesh around micropillar

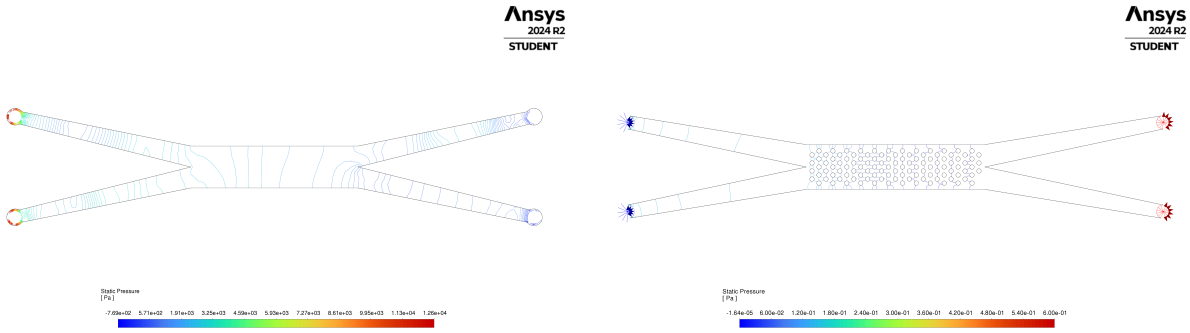
Figure 1: Meshes and geometries used in analysis

Using Fluent analysis system, I defined a species mixture material of liquid water and argon from predefined library of materials. In Inlet 1 the mass fraction was 99:1. Inlet 2 was only water. Both inlets were set as mass flow inlets with mass flow rate of $4.985 \cdot 10^{-5}\ \mathrm{kg} \cdot \mathrm{s}^{-1}$. Calculated velocity maps are shown in Figure 2, pressure maps in Figure 3 and argon concentration maps in Figure 4.



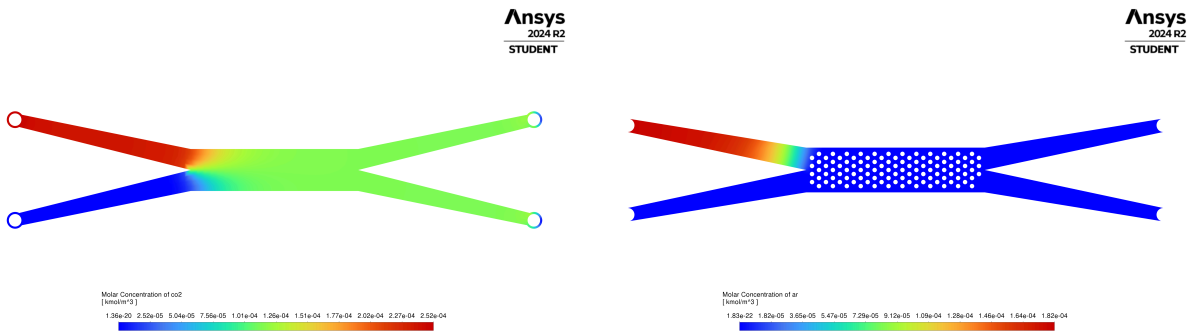
(a) without micropillars (b) with micropillars

Figure 2: Calculated velocity maps



(a) without micropillars (b) with micropillars

Figure 3: Calculated pressure maps



(a) without micropillars (b) with micropillars

Figure 4: Calculated velocity maps

3. Results

The primary objective was successful. I used FEM simulation to analyze species transport in reactor with and without micropillars. The secondary objective was a partial success. I used simulation software, but it was Ansys, not Comsol.